

Real Numbers: Exercises

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June 30, 2026

Exercise 1

Let $A, B, C, \dots \subseteq X$. The complement of A in X is

$$A^c = X \setminus A.$$

(a) Prove that $(A^c)^c = A$.

(b) Prove De Morgan's Laws:

$$(A \cap B)^c = A^c \cup B^c,$$

and deduce

$$(A \cup B)^c = A^c \cap B^c.$$

(c) Draw Venn diagrams illustrating these identities.

(d) Generalize these laws to arbitrary finite collections of sets.

Exercise 2

(a) Why does 4 divide every even integer square?

(b) Why does 8 divide every even integer cube?

(c) Why can 8 never divide twice an odd cube?

(d) Prove that the cube root of 2 is irrational.

Exercise 3

Let $x = A \mid B$, $x' = A' \mid B'$ be cuts in $\mathbb{Q}$. We defined

$$x + x' = (A + A') \mid \text{rest of } \mathbb{Q}.$$

(a) Show that although $B + B'$ is disjoint from $A + A'$, it may happen in degenerate cases that $\mathbb{Q}$ is not the union of $A + A'$ and $B + B'$.

(b) Infer that the definition of $x + x'$ as $(A + A') \mid (B + B')$ would be incorrect.

(c) Why did we not define $x \cdot x' = (A + A') \mid \text{rest of } \mathbb{Q}$?

Exercise 4

Prove that there exists no smallest positive real number. Does there exist a smallest positive rational number? Given a real number x , does there exist a smallest real number $y > x$?

Exercise 5

Given $x > 0$ and $n \in \mathbb{N}$, and $\epsilon > 0$, show that for some $\delta > 0$, if $u \in \mathbb{R}$ and $|u - y| < \delta$ then $|u^n - y^n| < \epsilon$

Exercise 6

Prove that real numbers correspond bijectively to decimal expansions not terminating in an infinite string of nines, as follows. The decimal expansion of $x \in \mathbb{R}$ is $N.x_1x_2\dots$, where N is the largest integer $\leq x$, x_1 is the largest integer $\leq 10(x - N)$, x_2 is the largest integer $\leq 100(x - (N + x_1/10))$, and so on.

- (a) Show that each x_k is a digit between 0 and 9.
- (b) Show that for each k there is an $\ell \geq k$ such that $x_\ell \neq 9$.
- (c) Conversely, show that for each expansion $N.x_1x_2\dots$ not terminating in an infinite string of nines, the set

$$\{N, N + \frac{x_1}{10}, N + \frac{x_1}{10} + \frac{x_2}{100}, \dots\}$$

is bounded and its least upper bound is a real number x with decimal expansion $N.x_1x_2\dots$

- (d) Repeat the exercise with a general base in place of 10.

Exercise 7

Let $f : A \rightarrow B$ be a function. That is, f is some rule or device which when presented with any element $a \in A$, produces an element $b = f(a)$ of B . The graph of f is the set S of all pairs $(a, b) \in A \times B$ such that $b = f(a)$.

- (a) If you are given a subset $S \subset A \times B$, how can you tell if it is the graph of some function? (That is, what are the set theoretic properties of a graph?)
- (b) Let $g : B \rightarrow C$ be a second function and consider the composed function $g \circ f : A \rightarrow C$. Assume that $A = B = C = [0, 1]$, draw $A \times B \times C$ as the unit cube in 3-space and try to relate the graphs of f, g , and $g \circ f$ in the cube.

Exercise 8

A rational number p/q is dyadic if q is a power of 2, $q = 2^k$ for some nonnegative integer k . A dyadic interval is $[a, b]$ where $a = p/2^k$ and $b = (p + 1)/2^k$. A dyadic cube is the product of dyadic intervals having equal length. The set of dyadic rationals may be denoted as $\mathbb{Q}_2$ and the dyadic lattice as $\mathbb{Q}_2^m$.

- (a) Prove that any two dyadic squares of the same size are either identical, intersect along a common edge, intersect at a common vertex, or do not intersect at all.
- (b) Show that the corresponding intersection property is true for dyadic cubes in $\mathbb{R}^m$.

Exercise 9

Given a cube in $\mathbb{R}^m$, what is the largest ball it contains? Given a ball in $\mathbb{R}^m$, what is the largest cube it contains?

Exercise 10

Let $b(R)$ and $s(R)$ be the number of integer unit cubes in $\mathbb{R}^m$ that intersect the ball and sphere of radius R , centered at the origin.

(a) Let $m = 2$ and calculate the limits

$$\lim_{R \rightarrow \infty} \frac{s(R)}{b(R)} \text{ and } \lim_{R \rightarrow \infty} \frac{s(R)^2}{b(R)}$$

(b) Take $m \geq 3$. What exponent k makes the limit interesting?

$$\lim_{R \rightarrow \infty} \frac{s(R)^k}{b(R)}$$

(c) Let $c(R)$ be the number of integer unit cubes that are contained in the ball of radius R , centered at the origin. Calculate

$$\lim_{R \rightarrow \infty} \frac{c(R)}{b(R)}$$

(d) Shift the ball to a new, arbitrary center and recalculate the limits.

Solutions to the Exercises

Solution 1

Part (a).

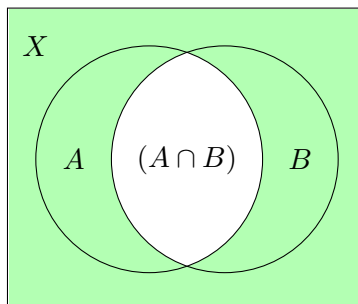
A^c is defined as $X \setminus A$. Let $x \in A^c$. This implies $x \notin A$. The complement of A^c would be $(A^c)^c$, which would leave us with $x \notin (A^c)^c$, which is equivalent to $x \notin A$. Therefore, we have shown that $(A^c)^c = A$.

Part (b).

Let $x \in A \cap B$. The complement of this would be $X \setminus (A \cap B)$, which would leave us with $x \notin (A \cap B)^c$. This means that $x \notin A$ or $x \notin B$, which is equivalent to $x \in A^c$ or $x \in B^c$. Therefore, we have shown that

$$(A \cap B)^c = A^c \cup B^c.$$

Part (c).



Part (d).

Let's consider n sets, and trying to generalize De Morgan's Laws. Let's assume these n sets are $A_i \subset X$, where $i = 1, 2, \dots, n$.

We can generalize De Morgan's Laws as follows:

$$\bigcup_{i=1}^n A_i^c = \left(\bigcap_{i=1}^n A_i \right)^c \iff \bigcap_{i=1}^n A_i^c = \left(\bigcup_{i=1}^n A_i \right)^c.$$

Solution 2

Proof. Part(a): An even integer can be represented as $2n$, where n is an integer. Knowing this, we can square $2n$, and we arrive at $4n^2$, which is divisible by 4. $\square$

Proof. Part(b): An even integer can be represented as $2n$, where $n \in \mathbb{Z}$. Cubing $2n$, we arrive at $8n^3$, which is divisible by 8. $\square$

Proof. Part(c): An odd integer can be represented as $2n + 1$, where $n \in \mathbb{Z}$. Cubing $2n + 1$, we arrive at $8n^3 + 12n^2 + 6n + 1$, which cannot be divided by 8 due to the remainder 1. $\square$

Proof. Part (d): Rational numbers are represented by $p/q, p, q \in \mathbb{Z}, q \neq 0$. Let's assume, for contradiction, that $\sqrt[3]{2}$ is rational, so $\sqrt[3]{2} = \frac{p}{q}$. Algebraic manipulation gives us $2q^3 = p^3$. Since the LHS is even, then the RHS must also be even, which implies that p is even, since odd integers cubed only produce odd integers, and even integers cubed only produce even integers. Thus, we let $p = 2k$ for some integer k , and we arrive at $2q^3 = 8k^3 \rightarrow q^3 = 4k^3$. Now, since the RHS is even, the LHS must also be even, which implies that q is even. This is a contradiction. Both p and q contain a factor of 2 which is divisible, thus showing that the rational expression was not in its simplest form. Therefore, $\sqrt[3]{2}$ is irrational. $\square$

Solution 3**Solution 4****Solution 5****Solution 6**

Proof. Part(a): $\square$

Proof. Part(b): $\square$

Solution 7**Solution 8****Solution 9****Solution 10**

Proof. $\lim_{R \rightarrow \infty} \frac{s(R)}{b(R)} = 0, \lim_{R \rightarrow \infty} \frac{s(R)^2}{b(R)} = \frac{64}{\pi}$ $\square$